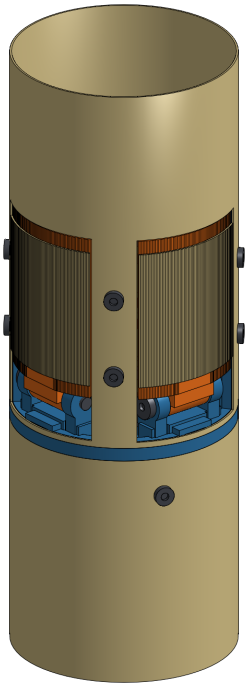
V3 – SG90 straight on the rotor: assembly guide

Built from the 22 mm V2. The spring cartridge, trigger, pushrod, yoke, cradle and all the layers between the servo and the rotor are gone: one SG90 hangs under the rotor and turns it through its own horn. The fins, cams, rotor-mounted latch, housing and tube slits are the V2 ones.

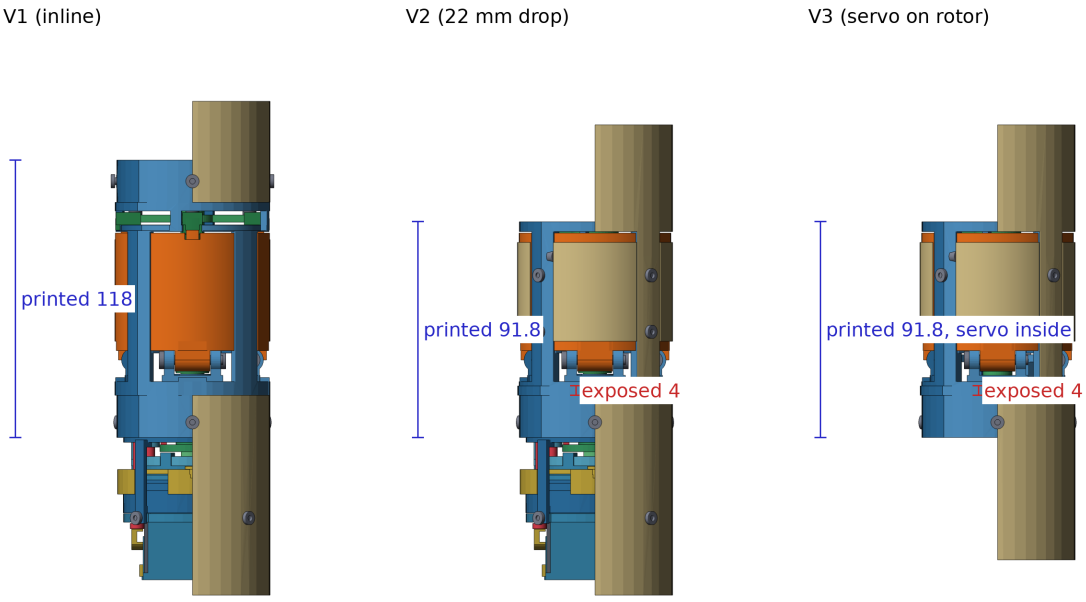
Onshape: ARC BT80 Airbrake - V1 Screwed + V2 Compact (Claude build). Elements: **V3-A** closed, **V3-B** fully open, **ASM-V3** assembly. Feature Studio: module mode *3* with *V2 compact head* on.



V3 closed. Same upper tube and slits as V2; the servo is inside the housing, flush with its bottom edge.

1 What changed

Item	V1	V2 (22 mm drop)	V3 (servo on rotor)
Printed housing incl. sleeve	118 mm	91.8 mm	91.8 mm (same housing)
Everything below the housing	cartridge, dock, cradle (to z –93.5)	same	**nothing**: the SG90 sits inside the housing, bottom flush at z –33
Whole assembly, top to bottom	178.5 mm	152.3 mm	**91.8 mm** (servo inside the housing)
Printed parts (solid, PLA 1.24 g/cm³)	180 g	169 g	**106 g** (–37 % vs V2)
Printed part count	19 types	19 types	**5 types**: 2 housing halves, 4 fins, 4 catch caps, rotor
Drive	wound spiral spring + SG90 trigger	same	SG90 turns the rotor 0 → 65°
Latch	tab and hook on the rotor	lug + finger on the rotor	same as V2

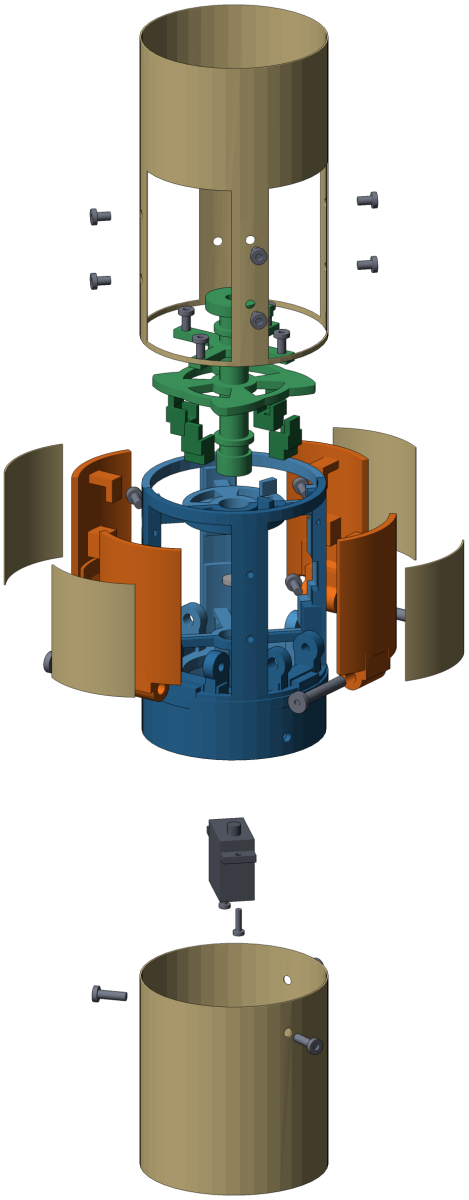


V1, V2 and V3 side by side (tubes cut in half). V3 keeps the V2 head; everything under it is replaced by the servo.

Removed printed parts: dock D, casing M, lid F, output hub U, bayonet collar N, locking pin I, winding cap T, double-D adapter X, thrust collar Q, pushrod A, release yoke B, servo cradle S, keeper G and horn saddle J (about 64 g), plus the spiral spring and return leaf.

2 How it works now

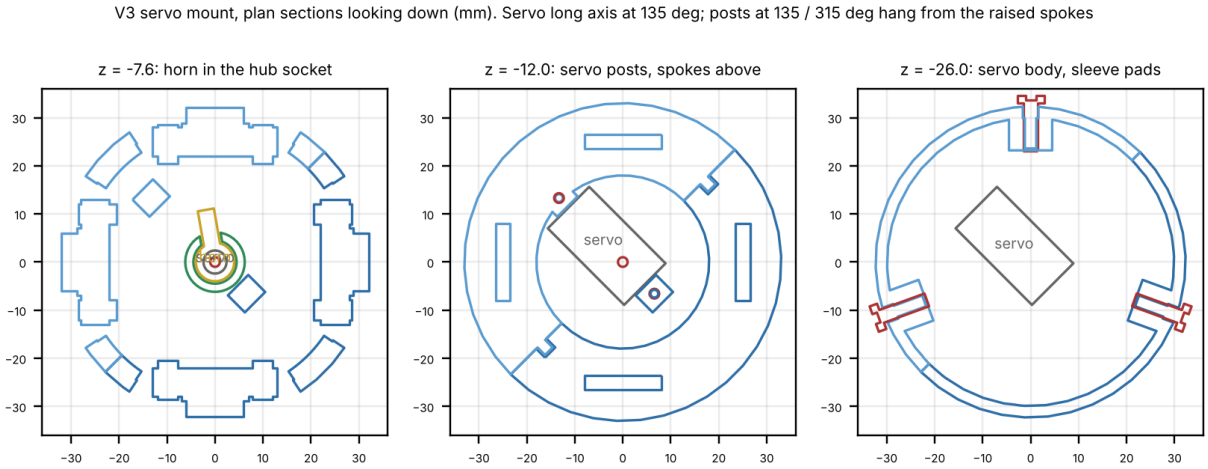
The servo horn sits in a socket in the underside of the rotor hub, so rotating the servo rotates the rotor. At 0° the finger on each catch cap holds the fin lug (the V2 latch). The latch lets go between 14° and 15° of rotor travel; the cams first touch the fins at about 26° and finish the kick at 65°. With the rotor at 65° the cam holds each fin open about 14° and the airflow carries it the rest of the way to the 90° hard stop, as in V1 and V2. The servo only has to move the rotor and slide the finger off the lug: it does not carry the airbrake drag, because at 90° the fins rest on the stop.



Exploded V3. Top to bottom: upper tube with its screws, horn screw, rotor with catch caps and tube cut-outs, housing with fins, horn, servo with its two flange screws, lower tube with sleeve screws.

3 Servo mount

The SG90 sits **inside the housing with its bottom flush with the housing bottom edge (z −33)**, so nothing sticks out below. It hangs by its flange from two posts (azimuth 135° and 315°, off the seam) that grow out of the lower spokes: M2×8 self-tapping screws go up through the two tab holes into 1.6 mm pilots in the posts. To make room, the rotor's lower bearing, its thrust collar and the four spokes that carry it moved **up** above the horn (bearing now z −0.8…2.2, spokes out to the support strips), and the rotor bottom stops at z −8.7. The shaft sits on the rotor axis, the horn lies in a round pocket and arm slot in the rotor hub, and the horn screw is dropped down through the rotor bore. The base ring has a 0.7 mm relief where the servo tab meets it. Three pads on the inside of the lower sleeve take the three M3×10 screws that hold the lower tube.



Plan sections: left through the horn socket (z –7.6), middle through the two servo posts (z –12), right through the servo body and the sleeve pads (z –26).

The servo, horn and tab-hole positions are standard SG90 numbers (body 22.7 × 12.2 × 22.7, tab holes 28.1 mm apart, horn arm 3.4 mm wide, 1.8 mm thick). **Measure your own servo and horn before printing**; the post pilots and the horn pocket are the two places that must match.

4 Torque check (SG90)

The load that matters is friction at the latch while the wind pulls the closed fins outward: the finger has to slide off the lug under that load. Fin area 1785 mm², latch 45.8 mm and skin centre 29 mm above the hinge, friction 0.2–0.4, finger at R24.8. The SG90 stall torque is 0.176 N·m at 4.8 V (0.216 at 6 V); I would only rely on about half of it (0.09 N·m). Rotor spin-up and the cam kick are negligible (about 0.001 N·m): the wind assists the opening.

Speed	Typical wind suction (Cp -0.2)	Pessimistic (Cp -0.6)
50 m/s	0.3 N on the finger, 0.006–0.012 N·m	0.9 N on the finger, 0.018–0.037 N·m
100 m/s	1.2 N on the finger, 0.025–0.049 N·m	3.7 N on the finger, 0.074–0.148 N·m
150 m/s	2.8 N on the finger, 0.055–0.111 N·m	8.4 N on the finger, 0.166–0.333 N·m
200 m/s	5.0 N on the finger, 0.099–0.197 N·m	14.9 N on the finger, 0.296–0.592 N·m

So the stock SG90 has margin to roughly **77–109 m/s** if the wind suction on the closed fins is as bad as the pessimistic case, and to roughly **134–188 m/s** in the typical case. Above that, use a metal-gear servo (about 0.22–0.25 N·m), polish the finger and lug, or add a ramp. The suction coefficient is an assumption: it needs a wind-tunnel or CFD number, or a ground test of the closed fins in a fan. The script is `tools/torque_check.py`.

5 Print list and hardware

File	Qty	Part	Notes
C1_housing_half_1_x1 / C2_housing_half_2_x1	1 + 1	Housing halves	Servo posts, sleeve pads and strips are part of the halves. Same print orientation as V2.
P_fin_x4	4	Fin	Same as V2.
L_catch_cap_x4	4	Catch cap	Same as V2.
R_rotor_x1	1	Rotor with horn socket	Hub with horn pocket and arm slot on the underside; the pocket opens into the bore (horn-screw access). Print upright.

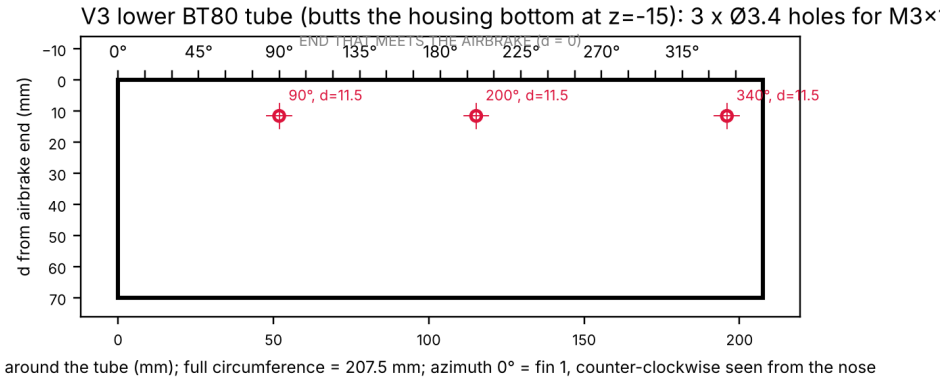
Ref in CAD	Fastener	Qty	Joint	Driver
HW seam screw 1-4	M3 × 8 socket-head cap	4	C2 counterbore → C1, z=8 and z=44	2.5 mm hex
HW L1-L4 cap screw	M3 × 8 button head	4	Rotor arm → catch cap boss	2.0 mm hex
HW H1-H4 hinge screw	M4 × 25 button head	4	Fin hinge, threads into the +Y ear	2.5 mm hex
HW top-tube support screw 1-8	M3 × 5 button head	8	Upper BT80 tube → strips between the fins	2.0 mm hex
HW sleeve screw 1-3	M3 × 10 button head	3	Lower BT80 tube → sleeve → pad (z=-26.5)	2.0 mm hex
HW servo flange screw 1-2	M2 × 8 button head, self-tapping	2	Servo tabs → housing posts	1.5 mm hex or PH0

Ref in CAD	Fastener	Qty	Joint	Driver
HW horn screw	M2 × 6 (the one supplied with the servo)	1	Horn → servo spline, from above through the rotor bore	PH0 / 1.5 mm hex

Totals: M3×5 × 8, M3×8 × 8, M3×10 × 3, M4×25 × 4, M2×8 × 2 plus the horn screw = 26 screws. Not printed or screwed: 1 SG90 servo with horn, 2 BT80 airframe tubes, epoxy for the tube cut-outs.

6 Tubes

The upper tube is cut exactly as in the V2 guide (four windows 38.6 × 63.8 mm, four 37.5 × 42 mm cut-outs glued to the fins, eight M3 holes). The lower tube needs only three 3.4 mm holes.



Lower tube template.

7 Assembly sequence

- Steps 1–4 as in V2: hinge screws, catch caps (M3×8) on the rotor arms, rotor into one half, fins, second half, four seam screws (2.5 mm hex).
- **Set the servo to its 0° position first** (servo tester or the flight computer), then press the horn on the spline so that the arm points where the pocket slot is when the rotor is locked at 0°.
- Drop the horn into the socket in the rotor hub from below, arm in its slot; slide the servo spline into the horn, flange under the two posts, long axis along the 135°/315° line.
- Drive the two M2×8 flange screws up through the tab holes. Then put the M2×6 horn screw down through the rotor bore and access hole (1.5 mm hex or small Phillips).
- Route the servo lead down the lower tube. Slide the lower tube over the sleeve and fit the three M3×10 screws.
- Glue the tube cut-outs to the fins, slide the upper tube on and fit the eight M3×5 screws, as in V2.
- Function check with the servo only: command 0° → 70° and back; the fins should unlatch around 15°, be kicked from about 26° and stop at 65° of horn travel.

8 Checks run on this model

Check	Result
V3-A closed / V3-B open (90°, 65°)	45 bodies, all watertight; no printed-part overlap; the only overlap among references is the servo spline inside the horn (intended)
Rotor, horn and catch caps turned 0 → 65° (5° steps)	clear of both housing halves and the servo at every step
Latch hold / release	fin blocked from 1.5–2° opening at rotor 0°; free from rotor 15° (14° still overlaps by 0.4 mm³)
Cam timing	cams first touch the closed fin at about 26° of rotor travel
Fin + glued cut-out swept 0.5 → 90° against the upper tube	clear at every angle
Seam screw access	straight 2.5 mm hex key reaches all four heads with the fins open
Assembly mates	21 features (fin groups, four hinges, rotor revolute): all solve OK

Open points for you: (1) Not tested on a print: the latch and the horn pocket. (2) The servo and horn dimensions are assumed. (3) The wind-suction number behind the torque check is an assumption; give me your top speed. (4) Mate motion is not scripted: drag the rotor and a fin in ASM-V3. (5) After the cam kick the fins stay out only while air flows; with no airflow they can fall back to about 14°. A positive hold open at 90° needs a detent or a longer cam. (6) Ground-impact strength is not analysed: that is the FEA you mentioned (the posts, sleeve pads and fin hinges are the first places to look).